

GEOGRAPHY

INDIAN GEOGRAPHY



INDIAN CLIMATE

MODULE - 6

INDIAN MONSOON

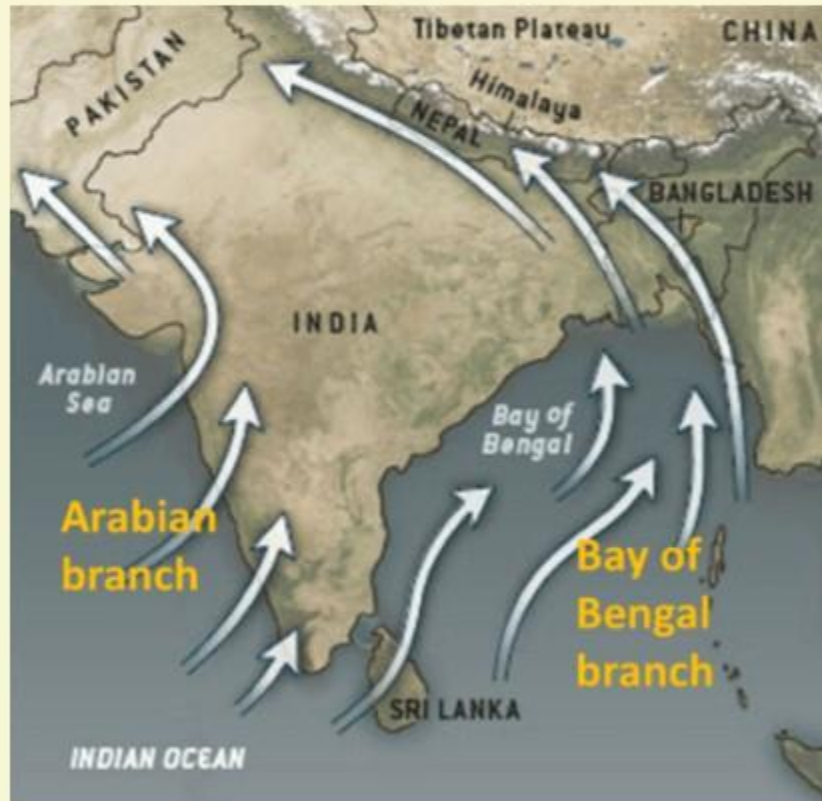
- **THE RHYTHM OF SEASONS**

THE RHYTHM OF SEASONS

3

SOUTHWEST MONSOON SEASON

As seen in last module,



11 As these winds approach the land, their southwesterly direction is modified by the relief and thermal low pressure over the northwest India.

12 The monsoon approaches the landmass in two branches:
(i) The Arabian Sea branch
(ii) The Bay of Bengal branch.

THE RHYTHM OF SEASONS

3

SOUTHWEST MONSOON SEASON

BRANCH NO. 1

MONSOON WINDS OF ARABIAN SEA

The monsoon winds originating over the Arabian Sea further split into three branches

1 Its one branch is obstructed by the Western Ghats.

2 These winds climb the slopes of the Western Ghats from 900-1200 m.

3 Soon, they become cool, and as a result, the windward side of the Sahayadris and Western Coastal Plain receive very heavy rainfall ranging between 250 cm and 400 cm.

4 After crossing the Western Ghats, these winds descend and get heated up. This reduces humidity in the winds.

5 As a result, these winds cause little rainfall east of the Western Ghats. This region of low rainfall is known as the rain-shadow area.

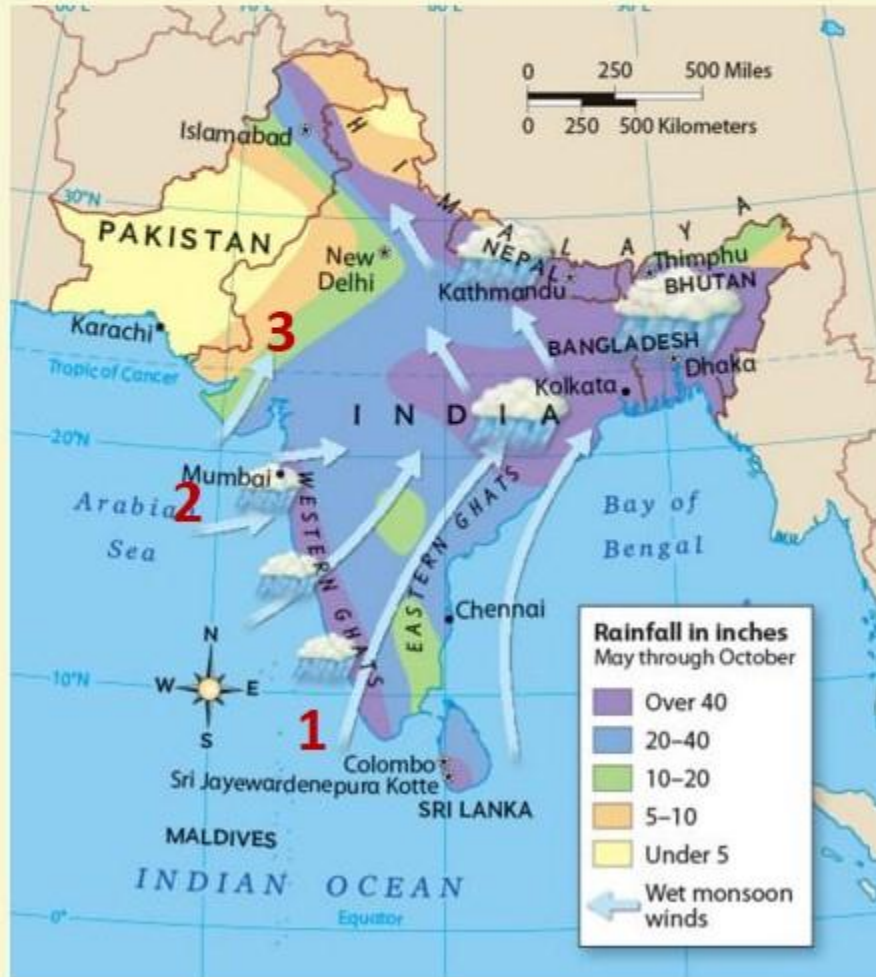
THE RHYTHM OF SEASONS

3

SOUTHWEST MONSOON SEASON

BRANCH NO. 2

MONSOON WINDS OF ARABIAN SEA



1

Another branch of the Arabian sea monsoon strikes the coast north of Mumbai.

2

Moving along the Narmada and Tapi river valleys, these winds cause rainfall in extensive areas of central India.

3

The Chotanagpur plateau gets 15 cm rainfall from this part of the branch. Thereafter, they enter the Ganga plains and mingle with the Bay of Bengal branch.

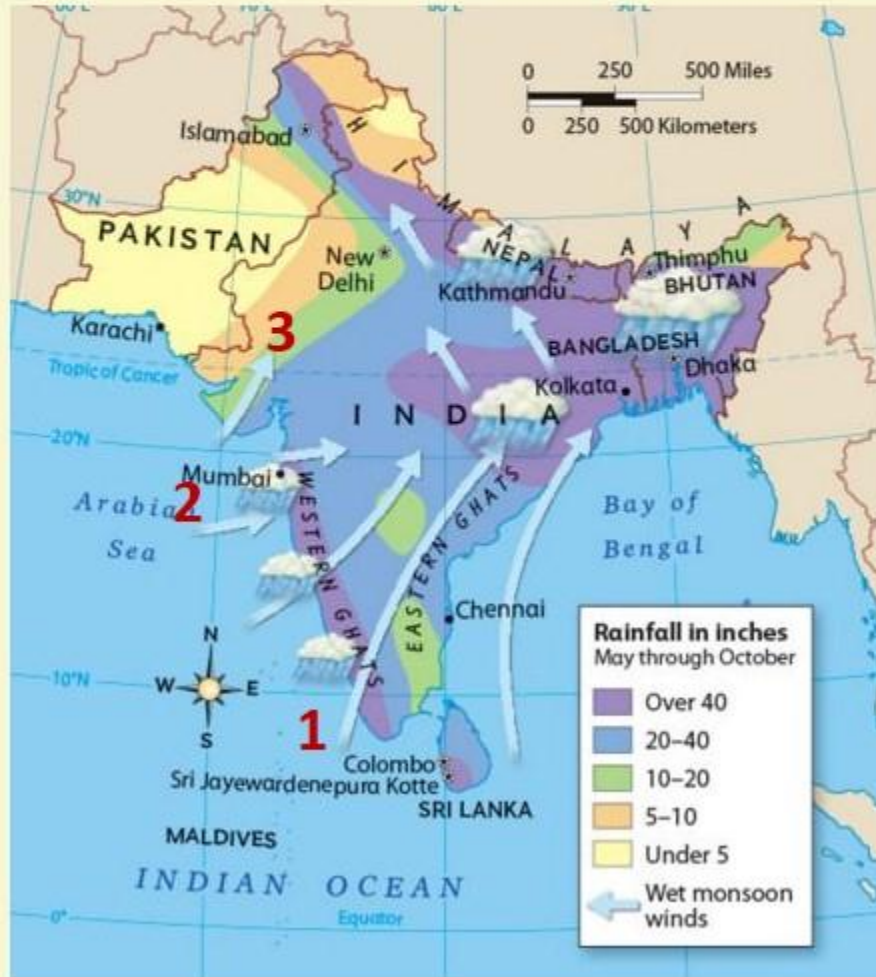
THE RHYTHM OF SEASONS

3

SOUTHWEST MONSOON SEASON

BRANCH NO. 3

MONSOON WINDS OF ARABIAN SEA



- 1 A third branch of this monsoon wind strikes the Saurashtra Peninsula and the Kachchh.
- 2 It then passes over west Rajasthan and along the Aravallis, causing only a scanty rainfall.
- 3 In Punjab and Haryana, it too joins the Bay of Bengal branch. These two branches, reinforced by each other, cause rains in the western Himalayas.

THE RHYTHM OF SEASONS

3

SOUTHWEST MONSOON SEASON

MONSOON WINDS OF BAY OF BENGAL



1

The Bay of Bengal branch strikes the coast of Myanmar and part of southeast Bangladesh. But the Arakan Hills along the coast of Myanmar deflect a big portion of this branch towards the Indian subcontinent.

2

Because of this, the direction of the wind shifts from south-westerly to easterly and south-easterly.

THE RHYTHM OF SEASONS

3

SOUTHWEST MONSOON SEASON

MONSOON WINDS OF BAY OF BENGAL



3

From here, this branch splits into two under the influence of the Himalayas and the thermal low is northwest India.

4

Its one branch moves westward along the Ganga plains reaching as far as the Punjab plains.

5

The other branch moves up the Brahmaputra valley in the north and the northeast, causing widespread rains. Its sub-branch strikes the Garo and Khasi hills of Meghalaya.

6

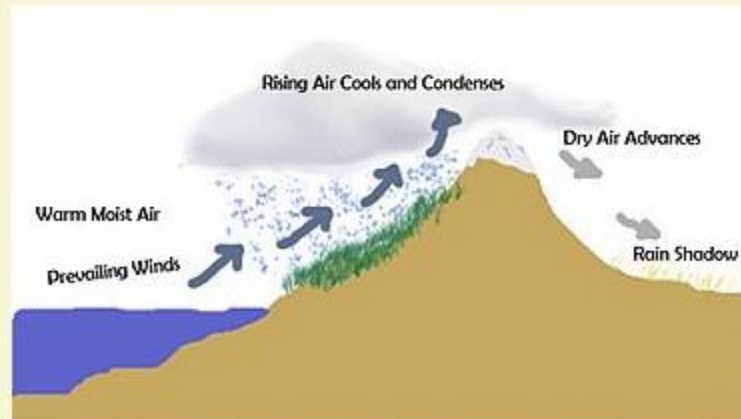
Mawsynram, located on the crest of Khasi hills, receives the highest average annual rainfall in the world.

THE RHYTHM OF SEASONS

3

SOUTHWEST MONSOON SEASON

MONSOON WINDS OF BAY OF BENGAL



7

Here it is important to know why the **Tamil Nadu coast remains dry during this season**. There are two factors responsible for it :

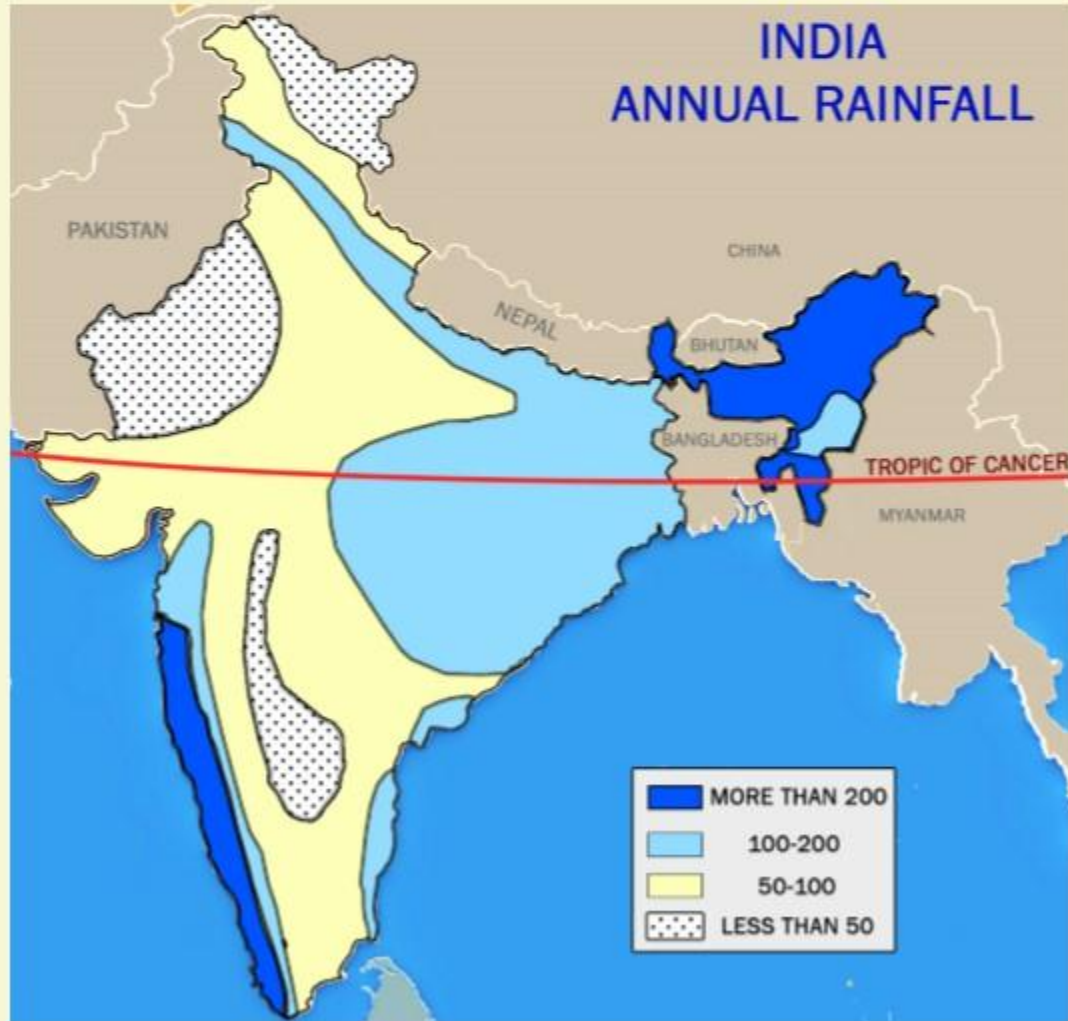
- i. The Tamil Nadu coast is situated parallel to the Bay of Bengal branch of southwest monsoon.
- ii. It lies in the rainshadow area of the Arabian Sea branch of the south-west monsoon.

THE RHYTHM OF SEASONS

3

SOUTHWEST MONSOON SEASON

CHARACTERISTICS OF MONSOONAL RAINFALL



1

Rainfall received from the southwest monsoons is **seasonal in character**, which occurs between **June and September**.

2

Monsoonal rainfall is largely governed by **relief or topography**. For instance the windward side of the Western Ghats register a rainfall of over 250 cm.

3

Again, the heavy rainfall in the northeastern states can be attributed to their hill ranges and the Eastern Himalayas.

THE RHYTHM OF SEASONS

3

SOUTHWEST MONSOON SEASON

CHARACTERISTICS OF MONSOONAL RAINFALL

4

The monsoon rainfall has a **declining trend with increasing distance from the sea**. Kolkata receives 119 cm during the southwest monsoon period, Patna 105 cm, Allahabad 76 cm and Delhi 56 cm.

5

- The monsoon rains occur in wet spells of few days duration at a time.
- The **wet spells are interspersed with rainless interval known as 'breaks'**.
- These breaks in rainfall are related to the **cyclonic depressions mainly formed at the head of the Bay of Bengal, and their crossing into the mainland**. Besides the frequency and intensity of these depressions, the passage followed by them determines the spatial distribution of rainfall.

CHARACTERISTICS OF MONSOONAL RAINFALL

BREAKS AND DEPRESSIONS IN SOUTH WEST MONSOON SEASON

- During July and August, there are certain periods when the monsoons become weak. Rainfall practically ceases over the country outside the Himalayan belt and southeast peninsula. This is known as break in the monsoon. *Breaks are likely to occur during the second week of August and last for a week.*
- *A major part of the South West Monsoon rainfall is generated by depressions* [intense low pressure] originating in the Arabian Sea and Bay of Bengal. Some depressions develop over land also.
- About 3-4 depressions are formed per month from June to September.
- *Almost all of them are sucked inward through the deltas of great rivers* [They need moisture to be alive], the Ganga, the Mahanadi, the Godavari, the Krishna and the Cauvery and cause heavy rain in these areas. Most of the rainfall in central and northern parts of the country is caused by these depressions.
- Most of the depression originate to the west of 90° E in *Bay of Bengal and move in north-west direction*. The location of depressions strongly coincide with the latitudinal position of ITCZ.

THE RHYTHM OF SEASONS

3

SOUTHWEST MONSOON SEASON

CHARACTERISTICS OF MONSOONAL RAINFALL

6 The summer rainfall comes in a heavy downpour leading to considerable run off and soil erosion.

7 Monsoons play a pivotal role in the agrarian economy of India because over three-fourths of the total rain in the country is received during the southwest monsoon season.

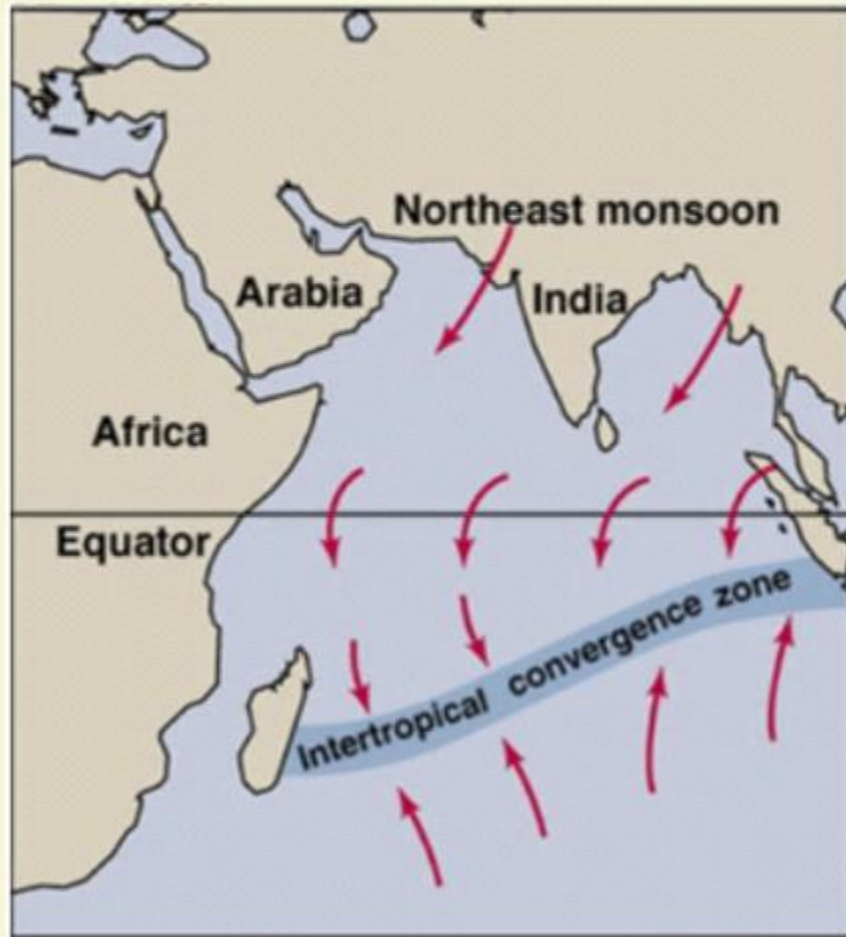
8 Its spatial distribution is also uneven which ranges from 12 cm to more than 250 cm.

9 The beginning of the rains sometimes is considerably delayed over the whole or a part of the country.

10 The rains sometimes end considerably earlier than usual, causing great damage to standing crops and making the sowing of winter crops difficult.

THE RHYTHM OF SEASONS

4 SEASON OF RETREATING MONSOON



NORTH-EAST MONSOON

- 1 The months of **October and November** are known for retreating monsoons.
- 2 By the end of September, the southwest monsoon becomes weak as the **low pressure trough of the Ganga plain starts moving southward** in response to the southward march of the sun.
- 3 The **monsoon retreats** from the western Rajasthan by the first week of September. It withdraws from Rajasthan, Gujarat, Western Ganga plain and the Central Highlands by the end of the month.

THE RHYTHM OF SEASONS

4 SEASON OF RETREATING MONSOON



- 4 By the beginning of October, the low pressure covers northern parts of the Bay of Bengal and by early November, it moves over Karnataka and Tamil Nadu.
- 5 By the middle of December, the centre of low pressure is completely removed from the Peninsula.
- 6 The retreating monsoon season is marked by clear skies and rise in temperature. The land is still moist.
- 7 Owing to the conditions of high temperature and humidity, the weather becomes rather oppressive. This is commonly known as the 'October heat'.

THE RHYTHM OF SEASONS

4 SEASON OF RETREATING MONSOON



- 8** In the second half of October, the mercury begins to fall rapidly, particularly in northern India.
- 9** The weather in the retreating monsoon is dry in north India but it is associated with rain in the eastern part of the Peninsula.
- 10** Here, October and November are the rainiest months of the year.
- 11** The widespread rain in this season is associated with the passage of cyclonic depressions which originate over the Andaman Sea and manage to cross the eastern coast of the southern Peninsula.

THE RHYTHM OF SEASONS

4 SEASON OF RETREATING MONSOON



12 These tropical cyclones are very destructive.

13 The thickly populated deltas of the Godavari, Krishna and Kaveri are their preferred targets.

14 Every year cyclones bring disaster here. A few cyclonic storms also strike the coast of West Bengal, Bangladesh and Myanmar.

15 A bulk of the rainfall of the Coromondal coast is derived from these depressions and cyclones. Such cyclonic storms are less frequent in the Arabian Sea.

WE EXPRESS OUR
SINCERE THANKS
FOR VIEWING THIS VIDEO

Presented by



Learning Space

For Suggestions:

suggestions@learningspace.in

To Contact us:

info@learningspace.in

OUR TEAM

G. V. Rao

K. Srikanth

D. Sunil Kumar

L. V. Krishna

Y. Deepthi

Amrita Naidu

M. Binukrishna

G. Ravi Babu

D. Joji

K. Victor Babu

Visit us at:

www.learningspacedigital.com

0 8 6 6 -2 4 4 4 4 7 2

0 9 8 4 9 9 4 2 2 9 9